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Question Bank (Unit 3: Quantum Physics)			
Subject	**Semiconductor and Quantum Physics**	**Subject Code**	**26SOST101**

SHORT QUESTIONS

1. Name the two phenomenon which were explained by quantum mechanics?
2. Explain de-Broglie's hypothesis?
3. Explain the concept of dual nature of matter?
4. Explain the wave function ψ?
5. Give the brief meaning of expectation value of position?
6. What is the physical significance of wave function?
7. Determine the de-Broglie wavelength of a photon.
8. What do you understand by a box in quantum mechanics?
9. Write the expression for Zero-point energy for a particle in one dimensional box. Why is it called zero-point energy?
10. Explain the Eigen values and Eigen function?
11. State any two limitations of classical mechanics.
12. Write the expression for the time-dependent Schrödinger wave equation.
13. Write the expression for the time-independent Schrödinger wave equation.
14. What do you understand by the quantum tunneling effect?
15. State one fundamental difference between classical computing and quantum computing.
16. What is a qubit in the context of quantum computing?
17. Write the formula relating the de Broglie wavelength of a particle to its momentum.
18. State the boundary conditions for the wave function of a particle confined in a one-dimensional box.
19. Give one practical application or example of the quantum tunneling effect.
20. Briefly explain the difference in the wave function's behavior at the boundaries of a finite potential well versus an infinite potential box.

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LONG QUESTIONS

1. Explain physical significance of Schrödinger wave function. Derive Schrodinger time independent wave equation.
2. Explain the normalized wave function. Derive Time-dependent Schrodinger equation.
3. Obtain expressions for the energy eigen values and normalized wave function for a particle confined in an infinitely deep potential well/one dimensional box.
4. What are de-Broglie waves? Show that the de-Broglie wavelength of a particle of momentum p is h/p. Also derive formula fo wavelength of particle in terms of kinetic energy.
5. Derive Schrodinger's time-independent equation for matter waves. Give a physical interpretation of a wave function.
6. Solve the Schrodinger wave equation for a particle in one dimensional rigid box of length a, having potential energy (V) as follows: V(x)= infinity for x<0 and x>a; V(x)=0 for 0 <= x<=a
7. What is tunneling effect in quantum mechanics. Also, explain the concept of quantum computing along with its application.
8. Discuss the concept of a wave function in quantum mechanics. Explain its physical significance and detail the essential mathematical properties (such as continuity, finiteness, single-valuedness, and normalization) that a wave function must satisfy to be considered physically acceptable.

Numerical Problems

1. Calculate the de-Broglie wavelength of an 'α' particle accelerated through a potential difference of 350 volt.
2. Calculate the de-Broglie wavelength of a proton accelerated through a potential difference of 350 volt.

3. Calculate the de-Broglie wavelength of a electron accelerated through a potential difference of 350 volt.
4. Calculate the kinetic energy of an electron if its de-Broglie wavelength equals the wavelength of sodium light. Given the wavelength of sodium light $\lambda = 5893$ Å.
5. An electron is bound in a one-dimensional potential box which has width $2.5 \times 10^{-10} m$. Assuming the height of the box to be infinite, calculate the lowest two permitted energy values of the electron.
6. Find the ground state energy of an electron moving in an infinitely high one-dimensional potential box of width 1 Å.
7. Calculate the de-Broglie wavelength of an electron whose kinetic energy is $50\ eV$.

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Question Bank

(Unit 4: Photonics and Optical Communication)

Subject	Semiconductor and Quantum Physics	Subject Code	26SOST101

SHORT QUESTIONS

1. Define spontaneous emission of radiation.
2. Define stimulated emission of radiation.
3. Distinguish between spontaneous and stimulated emission.
4. State the significance of Einstein's A and B coefficients.
5. Write the mathematical relation between Einstein's coefficients for absorption and stimulated emission.
6. Define a metastable state and state its importance in laser operation.
7. Define population inversion.
8. State the conditions necessary for achieving population inversion.
9. Explain the meaning of pumping in a laser system.
10. Define lasing action.
11. List the essential components of a laser.
12. Identify the active medium and pumping mechanism used in a Ruby laser.
13. Identify the active medium and pumping mechanism of a He-Ne laser.
14. State the principle of operation of a semiconductor laser.
15. Differentiate between three-level and four-level laser systems.
16. List any four applications of lasers.

17. State any two advantages of semiconductor lasers.
18. Mention two applications of He-Ne lasers.
19. Mention two applications of Ruby lasers.
20. Differentiate between ordinary light and laser light.
21. Describe the principle of optical fiber.
22. Briefly explain the acceptance angle in optical fiber.
23. Name the types of optical fibers.
24. Define the critical angle in optical fiber.
25. Explain the construction of optical fiber.
26. Write the expression for numerical aperture in terms of core and cladding refractive indices.
27. State the relationship between acceptance angle and numerical aperture.
28. Differentiate between step-index and graded-index optical fibers.
29. Differentiate between single-mode and multimode optical fibers.
30. List the major types of optical fibers.
31. State any four applications of optical fibers.
32. Mention two advantages of optical fiber communication.
33. Why is the cladding provided around the core of an optical fiber?

LONG QUESTIONS

1. Explain the working principle of a He-Ne laser with the help of a well-labelled energy level diagram.
2. What is population inversion? How is it achieved in a He-Ne laser? Explain why it is essential for laser action and how the presence of helium helps in achieving it in the He-Ne system.
3. Describe the construction and working of a Ruby Laser. Explain the energy level diagram involved.

4. What are Einstein's coefficients? Derive the relation between them and explain their significance in the working of a laser.
5. Explain the differences between step-index and graded-index optical fibres with the help of diagrams. How does light propagate differently in both types?
6. Explain the concept of numerical aperture in optical fibres. Derive an expression for numerical aperture and acceptance angle using Snell's law.
7. Describe single mode and multi-mode optical fibres. Explain the difference between them.

Numerical Problems

1. Calculate the energy of laser pulse in a ruby laser for 2.8x1019 Cr+3 ions. If the laser emits radiation of wavelength 6943Å.
2. Calculate the population ratio of two states in He-Ne Laser that produces light of wavelength 6000 Å at 300K.
3. Compute the numerical aperture and hence the acceptance angle for an optical fibre given those refractive indices of the core and the cladding are 1.42 and 1.40 respectively.
4. An optical fibre has the core refractive index n1=1.36 and relative difference in index =0.025. Find (i) refractive index of cladding (ii) numerical aperture (iii) acceptance angle.
5. An optical fibre has a numerical aperture of 0.20 and a cladding refractive index of 1.59. Determine the acceptance angle for the fibre in water which has a refractive index of 1.33.
6. An optical fibre has n1 = 1.55, n2 = 1.50 and core diameter = 50μm
7. Calculate (i) numerical aperture; (ii) acceptance angle; and (iii) how many reflections per meter are suffered by the guided ray at steepest angle with respect to fibre axis.
8. A glass fibre has a core material of refractive index 1.45, cladding material of refractive index 1.40. If it is surrounded by air, compute the critical angle
   i) at the core cladding boundary
   ii) cladding air boundary